VALOX™ 508R resin

PBT/PC–SABIC

Good Rigidity,Low warping,Heat Resistance, Good,Glass fiber reinforced

Introduction

30% 玻纤增强PBT + PC。 出色的机械和热性能。非阻燃。 降低翘曲特性。 应用与VALOX 420R相同

Product Description

Supplier	SABIC
Generic	PBT/PC
Material Status	Commercial: Active
Features	Good Rigidity,Low warping,Heat Resistance, Good,Glass fiber reinforced

Technical Data

MECHANICAL	Nominal value	Unit	Test method
tensile strength			
Fracture, Type I, 5 mm/min	112	Mpa	ASTM D638
Flexural Strength			
Fracture, 1.30 mm/min, 50 mm span	193	Mpa	ASTM D790
Flexural Modulus			
1.3 mm/min，50mm跨距	7030	Mpa	ASTM D790
Rockwell hardness	119		ASTM D785
IMPACT	**Nominal value**	**Unit**	**Test method**
Izod Notched Impact strength			
23℃	650	J/m	ASTM D4812
23℃	90	J/m	ASTM D256
-30℃	80	J/m	ASTM D256
Instrument impact energy			
23℃			
peak value	6.9	N.m	ASTM D 3763
total value	8.2	N.m	ASTM D 3763
THERMAL	**Nominal value**	**Unit**	**Test method**
HDT			
Unannealed, 6.4 mm			
0.45 MPa	215	℃	ASTM D 648
1.8MPa	176	℃	ASTM D 648
Coeff.of linear therm expansion			

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MD			
-40 to 40 ℃	2.34E-05	1/℃	ASTM E 831
60 to 138 °	1.62E-05	1/℃	ASTM E 831
Relative temperature index			
ELECTRICAL	125	℃	UL 746B
Impact mechanical performance	110	℃	UL 746B
Non mechanical impact performance	125	℃	UL 746B

PHYSICAL	Nominal value	Unit	Test method
Density	1.5	g/cm^3	ASTM D792
Specific product	0.66	cm^3/g	ASTM D 792
Water Absorption			
24hr	0.06	%	ASTM D570
Shrinkage			
MD			
1.5-3.2 mm, flowing	0.3-0.5	%	SABIC Method
3.2-4.6 mm, flowing	0.5-0.8	%	SABIC Method
1.5-3.2 mm，TD	0.4-0.6	%	SABIC Method
3.2-4.6 mm，TD	0.6-0.9	%	SABIC Method

ELECTRICAL	Nominal value	Unit	Test method
Volume resistivity	5.9E+16	ohms·cm	ASTM D257
Dielectric strength			
3.2 mm in air	23.6	kV/mm	ASTM D149
1.6 mm in oil	29.1	kV/mm	ASTM D149
Relative permittivity			
100 Hz	3.6		ASTM D150
1 MHz	3.6		ASTM D150
Dissipation factor			
100 Hz	0.0014		ASTM D150
1 MHz	0.02		ASTM D150
Arc resistance			
Tungsten wire	6	PLC Code	ASTM D495
Hot wire ignition {PLC}	1	PLC Code	UL 746A

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High voltage arc marking rate {PLC}	2	PLC Code	UL 746A
High current arc ignition (surface {PLC})	3	PLC Code	UL 746A
Comparative Tracking Index	4	PLC Code	UL 746A

FLAME CHARACTERISTICS	Nominal value	Unit	Test method
Flame retardant grade 94HB	1.47	mm	UL 94
ultraviolet radiation			
Water exposure/immersion	F2	-	UL 746C

Process Conditions

Injection molding	Nominal value	Unit	Test method
Drying temperature	120	°C	
Drying time	3-4	hrs	
Drying time (gradually increasing)	12	hrs	
Maximum water content	0.02	%	
Melt temperature	250-265	°C	
Nozzle temperature	245 - 260	°C	
Front temperature	250-265	°C	
Mid section temperature	245-260	°C	
Post stage temperature	240-255	°C	
Mold temperature	65-90	°C	
Back pressure	0.3-0.7	MPa	
Screw speed	50-80	rpm	
Shooting cylinder size	40-80	%	
Depth of exhaust groove	0.025-0.038	mm	